Republic Testnet Weekly Reward Model & Payout System

A fair, transparent, and anti-gaming reward system for GPU miners and job creators.

Each week, **1,600,000 RAI** is distributed using a **continuous multiplicative scoring model** that rewards:

- Real GPU processing
- Reliable job creation
- Consistent uptime
- Temporal coverage across the week
- Validator participation

This system removes reward cliffs and replaces them with **smooth contribution-based allocation**.

Overview

Traditional tiered mining systems create problems:

Problem	Effect
Hard thresholds	4,999 jobs ≠ 5,000 jobs
Burst mining	Short-term optimization instead of stability
Job spam	Low-quality submissions
Sybil attacks	Wallet splitting advantages

Republic Testnet replaces tiers with a **continuous contribution function**:

```
Final Score = Effort × JobCreation × Presence × Steadiness × Helpfulness × BuilderBonus
```

Your **Final Score** determines your share of the weekly reward pool.

Weekly Reward Pool

```
Total Weekly Distribution = 1,600,000 RAI
```

```
Maximum Per Wallet = 240,000 RAI (15% cap)
```

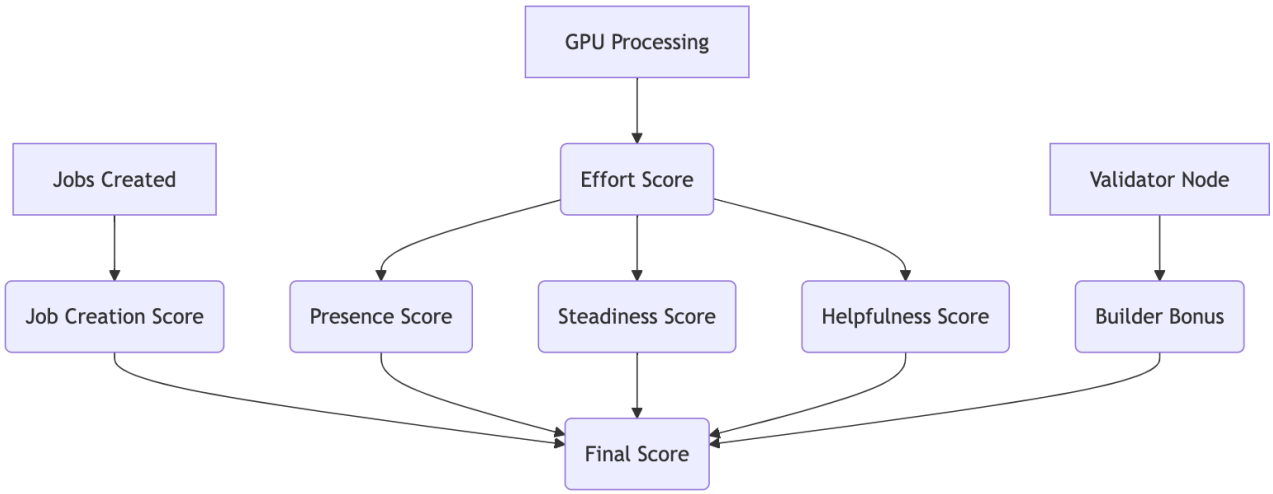
Rewards are distributed **proportionally**:

```
wallet_reward = (wallet_score / total_network_score) × 1,600,000
```

Then capped at:

```
min(wallet_reward, 240,000)
```

Contribution Model Architecture



Metric Glossary

Abbrev	Full Meaning
Sub	Jobs Submitted: Number of jobs this wallet created
CrC	Creation Completed: Number of submitted jobs that reached COMPLETED status
EfC	Effort Completed: Jobs processed on GPU with result submitted
SR	Success Rate: Completed ÷ Attempted jobs as processor
JC	Job Creation Score (ranges from 1.00 to 1.08)
Eff	Effort Score (0 if fewer than 5,000 processed jobs)
Pres	Presence Score (derived from processed-job active hours only)
Steady	Steadiness Score (day spread × streak factor across the week)

Abbrev	Full Meaning
Help	Helpfulness Score (hour spread × success rate, requires ≥ 5,000 jobs)
Build	Builder Bonus (validator uptime proxy + contribution share)
Final	Final Contribution Score (used to calculate weekly RAI allocation)

The Six Contribution Scores

Each wallet receives six independent scores. These are multiplied together to compute contribution weight

Score	Formula	Purpose
JC	`1 + min(0.08, completion_rate × 0.08)`	Rewards useful job creation
Eff	`sqrt(EfC) × SR`	Measures real GPU contribution
Pres	`(unique_active_hours / 168) × SR`	Rewards consistent uptime
Steady	`(unique_active_days / 7) x (long_streak_days)`	Rewards weekly activity distribution
Help	`1 + (unique_hours_of_day / 24) × 0.25 × SR`	Rewards 24-hour coverage
Build	`1 + (uptime_proxy × 0.05) + (contribution_share × 0.05)`	Rewards validator participation
Final	`Eff × JC × Pres × Steady × Help × Build`	Determines reward share

Reward Allocation Algorithm

Step 1:

```
calculate Final Score for each wallet
```

Step 2:

```
sum all wallet scores
```

Step 3:

```
distribute proportional rewards
```

Step 4:

```
apply 15% wallet cap
```

System Design Principles

The reward system enforces:

Principle	Mechanism
Anti-spam	Job Creation Score
Anti-burst mining	Steadiness Score
Anti-downtime	Presence Score
Anti-timezone gaps	Helpfulness Score
Anti-Sybil advantage	Multiplicative scoring
Infrastructure incentives	Builder Bonus

Script Workflow

The payout script:

1. Fetches job data from Yaci Explorer API
2. Caches responses locally
3. Converts wallet ↔ validator addresses automatically
4. Computes all six contribution scores
5. Allocates 1,600,000 RAI proportionally
6. Applies 240,000 cap
7. Outputs payout files

Running the Script

Git clone this repository:

```
https://github.com/geofvn/Republic-gpuminer-rewardcal.git
```

Install dependencies once:

```
pip install requests bech32 or pip install requirement.txt
```

Run:

```
python3 weekly_payout.py
```

Choose:

```
YES → custom interval
NO → last 7 days
```

Important Rules

```
1. Only COMPLETED jobs count
2. Effort requires ≥ 5,000 processed jobs
3. Helpfulness requires ≥ 5,000 processed jobs
4. Builder bonus requires bonded validator
5. Maximum payout per wallet = 240,000 RAI
```

Operational Recommendation

Run the payout script weekly:

```
Sunday or Monday UTC preferred
```

Ensures consistent reporting windows

Transparency Commitment

This reward model ensures:

- fairness
- predictability
- Sybil resistance
- validator incentives
- global compute availability

Designed for the Republic community 👥